

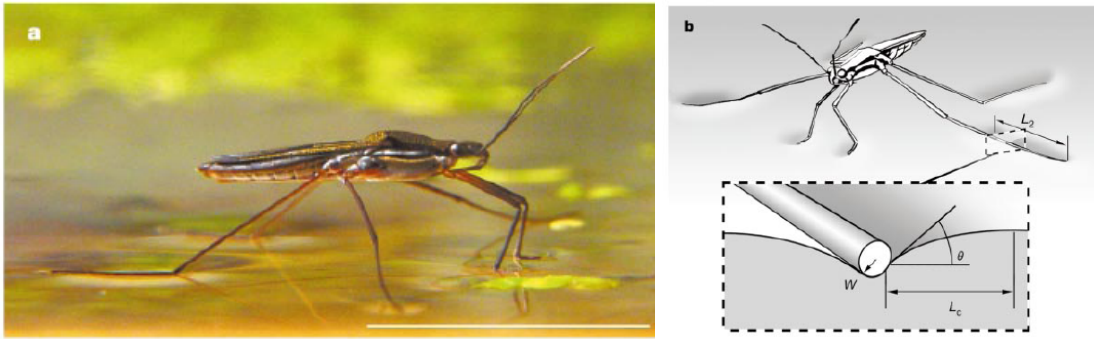
Homework 1 & 2:

Capillarity: Measuring Surface Tension, Minimal Surfaces and Laplace Pressure

(Original problem(s): do not repost anywhere without permission)

(Problem 1-3 HW 1, & problem 3 optional for UGs; Problem 4-6, HW 2, & prob 6 optional for UGs).

- (1) Watch these three videos. (a) <https://youtu.be/MUImkSnrAzM?si=Vjz7-Z9QHxcHST-N> (b) <https://youtu.be/XDQ21ngtpkI?si=T6HFs5dvztL1bKAe> (c) <https://youtu.be/QiZL-8tkMdo?si=bozLI6js3rMI4h73> Two questions: (i) Among the phenomena discussed in video (a), pick any two and run the experiment in your kitchen. Feel free to include links to videos you make, but more importantly, summarize the idea, the experiment, the observation, and how well your results compared to what was shown in the video. (ii) What is your favorite dimensionless number? (iii) list as many characteristic lengths as you can from the video, especially those with surface tension involved in the definition.
- (2) **Butterfly Proboscis:** Butterflies (and many other insects) sip nectar (blood, sap, water, *etc.*) using their proboscis, a straw-like capillary tube, and though they don't probably know this, interfacial phenomena play a crucial role in their survival. For argument's sake, we can estimate the radius of a long cylindrical capillary needed for bringing water to a height of 1 mm, 10 mm, and 100 mm without the need for external pumping. State any assumptions you make to calculate the estimated capillary radius. What is the hydrostatic pressure at 0.1 mm, 5 mm, and 50 mm under water? How does it compare to atmospheric pressure? Colloidal and Interfacial Phenomena in Proboscis could be a nice topic for a final project. For a more detailed discussion on microfluidics involved in proboscis, see the following reference:
http://jeb.biologists.org/content/217/12/2130?ijkey=5fc423a6dd6efe3313074666b54a85b53a1a603e&keytype=tf_ipsecsha
- (3) **Water Striders:** (a) An insect can walk on water as long as an additional vertical force counterbalances its weight W . In class, we discussed how floating objects deform the surface of a fluid, such as water, how to calculate the curvature of such a surface, and the Laplace pressure associated with curved interfaces. Using this knowledge, discuss how the maximum weight that can be supported depends upon the surface tension of the underlying fluid. What would be a good upper limit to the weight that can be supported by a cylindrical rod placed on water? [Water Strider & relevant Interfacial Phenomena could be another good theme for a final project].
(b) Water striders have thin hydrophobic legs, and penetrating the water surface requires a force equal to the surface tension of water times the perimeter of the leg. Carbon Nanotube forests, on the other hand, have multiple cylindrical rods that can together form a superhydrophobic surface. In general, for elastic rods, buckling occurs when the compressive force exceeds the Euler's critical load
$$F_{Euler} = \left(\pi/2\right)^2 ER^4/L^2$$
where E is the elastic modulus, R is the radius of the rod, and L is the length of the rod. What is the critical length of the rod beyond which buckling occurs under capillary forces? Compare this critical length to the capillary length of the fluid. Estimate the critical length for human hair ($E = 4$ GPa, $R = 40$ micron), assuming the hair is brought in contact with water.



- (4) “The Cheerios Effect”: <http://people.maths.ox.ac.uk/vella/Vella2005.pdf> Attraction between two plates dipped into a fluid depends upon interfacial deformation (and in turn on surface tension, contact angle, *etc.*) Re-derive the expressions (1)-(6) given in the paper titled “The Cheerios Effect”. State all the assumptions made, and any simplifications used or implied. Is there a characteristic separation distance beyond which no attraction or repulsion between plates is felt? Is there a dimensionless group that can be computed to estimate how strong or weak the capillary interaction forces can get for a given fluid? Will the attraction get stronger or weaker as the temperature of the fluid and the two plates are raised, such that no temperature gradients are created? (Advanced problem, for self-study only: Check if you can rederive the expressions outlined in this paper for attraction between spherical particles confined to the liquid/air interface).
- (5) **Show the derivation for the size of a puddle formed by a large drop on a solid substrate and of a floating lens formed on an immiscible liquid substrate, assuming the capillary length is exceeded in both cases.**
- (6) **“Can soft solids be shaped by capillary forces, just like ordinary liquids?”** (Required for graduate students, optional for undergraduates). Find and read Physical Review Letter 108, 094301, 2012. Write a critical review (one to two pages, Times New Roman, font size 12, single spacing) of the published paper without using AI help. (HINT: Summarize the key new ideas presented in this paper, check if the derivations are accurate, comment on any overly simplistic or unphysical assumptions made, and check their limits of applicability, list any papers the authors might have missed, and find any grammatical or mathematical errors herein. (Also see Soft Matter, 2012, 8, 7177, contact angle on soft surfaces, and related papers)).

Provide only simple estimates/ expressions in problems 2-4. Comments on missing details, or how a more accurate analysis can be carried out, are welcome. In problem 5, you can write a longer review if you have something exceptional to contribute. Among other things, heroic effort is up for grabs for exceptional effort.